

Question			Marking details	Marks available					Prac
				AO1	AO2	AO3	Total	Maths	
2	(a)	(i) (ii)	Mass of a molecule [accept: atom or particle] [of the gas] Mean [accept: average] square velocity / speed [of a molecule of the gas]	1 1			2		
	(b)		N_A , [Avogadro's number is] the number of molecules per mole [or in a mole] [of gas] (1) There are n moles [of the gas] (1) So the total number of molecules [in the gas] = $n N_A$ (1)	3			3		
	(c)		Realising that $KE = N_{(A)} \frac{1}{2} \overline{mc^2}$ (1) Substitution of $pV = nRT$ (1) Convincing algebra, e.g handling $\frac{2}{3} \frac{1}{3} \frac{3}{2}$ (1) Explanation that $n = 1$ [NB. if M used need to see molar mass when $n = 1$] (1) Alternative: $\frac{1}{2} \overline{mc^2} = \frac{3}{2} kT$ (1) Multiplication by $N \rightarrow \frac{1}{2} N \overline{mc^2} = \frac{3}{2} NkT$ (1) Justification that $R = kN_A$ (1) with justification (1)	4			4	2	

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	(d)		$n = \frac{pV}{RT} = \frac{115 \times 10^3 \times 2.2 \times 10^{-3}}{8.31 \times 294} = 0.104 \text{ [mol] (1)}$ Final temperature, $T = \frac{pV}{nR} = \frac{115 \times 10^3 \times 2.6 \times 10^{-3}}{0.104 \times 8.31} = 346.0 \text{ K (1)}$ [value depends upon rounding of n. No rounding $\rightarrow 347.5 \text{ K}$] $\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} \times 0.104 \times 8.31 (346 - 294) = 67.4 \text{ J (1)}$ Allow ecf on 1 mol for the last two marks. Alternative: $\Delta U = \frac{3}{2} (p_1 V_1 - p_2 V_2)$ or $\Delta U = \frac{3}{2} p \Delta V$ (for constant pressure) (1) $= \frac{3}{2} 115 \times 10^3 (2.60 - 2.20) \times 10^{-3} \text{ (1)}$ $= 69 \text{ J (1)}$		3		3	3	
			Question 2 total	9	3	0	12	5	0